# Where the Headroom Lies: Block $\ell_p$-Norm Statistics for Cyclic Join Bounds

Extended version: full proofs and additional experiments

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This note accompanies the ICDE 2027 submission and gives the proofs that the paper states in condensed form: the $\ell_p$ Composition Inequality (the validity of every LpH-CEG bound) and the Cyclic Improvement theorem, and the $bs = 1$ experiments on JOB, a TPC-H-like schema, data size and skew. Notation follows the paper.

## 1 The $\ell_p$ Composition Inequality

**Setup.** A hyperpath $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m)$ adds a block of relations $F_j$ at step $j$, $S_j = S_{j-1} \cup F_j$, and $S_m$ is the set of all relations. Write $A_j = \mathrm{attr}(F_j)$, $V_j = \mathrm{attr}(S_j)$ ($V_0 = \emptyset$, $V_m = V$), the interface $I_j = A_j \cap V_{j-1}$ and $E_j = A_j \setminus I_j$. Then

(F1) $E_j = V_j \setminus V_{j-1}$, so $\{E_j\}_j$ partitions $V$;

(F2) $I_j \subseteq V_{j-1}$.

Weights $w_{j,p} \geq 0$ ($p \in \mathcal{P} \subseteq [1, \infty]$) satisfy the attribute-wise cover constraint of the paper: for every attribute $a \in V$, bound at step $j(a)$ (i.e. $a \in E_{j(a)}$),

$$w_{j(a)} + \sum_{j' :\, a \in I_{j'}} s_{j'} \;\geq\; 1, \qquad w_j := \sum_p w_{j,p}, \quad s_j := \sum_p w_{j,p}/p \;\; (1/\infty = 0). \tag{C}$$

A *polymatroid* is $h : 2^V \to \mathbb{R}$ with $h(\emptyset) = 0$, monotone and submodular; $h(X \mid Y) := h(X \cup Y) - h(Y)$.

**Lemma 1** (Conditioning)**.** *For a polymatroid $h$ and $Y \subseteq Z$: $h(X \mid Y) \geq h(X \mid Z)$.*

*Proof.* Submodularity on $X \cup Y$ and $Z$ gives $h(X \cup Y) + h(Z) \geq h(X \cup Y \cup Z) + h((X \cup Y) \cap Z)$, and $h((X \cup Y) \cap Z) \geq h(Y)$ by monotonicity since $Y \subseteq (X \cup Y) \cap Z$. Rearrange. □

**Lemma 2** (Fractional subadditivity)**.** *Let $f$ be monotone submodular on subsets of a finite set with $f(\emptyset) = 0$, and let $\lambda_i \geq 0$ satisfy $\sum_{i:\, a \in T_i} \lambda_i \geq 1$ for every $a \in S$. Then $\sum_i \lambda_i f(T_i) \geq f(S)$.*

*Proof.* Order $S = (a_1, \ldots, a_k)$ and let $S_{<t} = \{a_1, \ldots, a_{t-1}\}$. By monotonicity, the chain rule and Lemma 1, $f(T_i) \geq f(T_i \cap S) = \sum_{a_t \in T_i} f(a_t \mid T_i \cap S_{<t}) \geq \sum_{a_t \in T_i} f(a_t \mid S_{<t})$. Weighting by $\lambda_i$ and exchanging sums, $\sum_i \lambda_i f(T_i) \geq \sum_t f(a_t \mid S_{<t}) \sum_{i:\, a_t \in T_i} \lambda_i \geq \sum_t f(a_t \mid S_{<t}) = f(S)$, using $f(a_t \mid S_{<t}) \geq 0$. □

**Lemma 3** (The obligation)**.** *For every polymatroid $h$ and every $w$ satisfying (C),*

$$\sum_j \Big[ s_j\, h(I_j) + w_j\, h(E_j \mid I_j) \Big] \;\geq\; h(V).$$

*Proof.* Let $g_j = h(E_j \mid V_{j-1})$, $\delta_j = \max(0, 1 - w_j) \in [0,1]$ and $f_j(S) = h(S \mid V_{j-1})$ for $S \subseteq E_j$; $f_j$ is monotone, submodular and $f_j(\emptyset) = 0$.

(i) For each $j'$, the chain rule along the partition (F1) and Lemma 1 give $h(I_{j'}) = \sum_j h(I_{j'} \cap E_j \mid I_{j'} \cap V_{j-1}) \geq \sum_j f_j(I_{j'} \cap E_j)$. Hence $\sum_{j'} s_{j'} h(I_{j'}) \geq \sum_j \sum_{j'} s_{j'} f_j(I_{j'} \cap E_j)$.

(ii) Fix $j$. If $\delta_j > 0$ then by (C) every $a \in E_j$ has $\sum_{j' : a \in I_{j'}} s_{j'} \geq 1 - w_j = \delta_j$, so the sets $I_{j'} \cap E_j$ with weights $s_{j'}/\delta_j$ fractionally cover $E_j$, and Lemma 2 gives $\sum_{j'} s_{j'} f_j(I_{j'} \cap E_j) \geq \delta_j\, g_j$. If $\delta_j = 0$ the same inequality holds because the left side is non-negative.

(iii) By (F2) and Lemma 1, $h(E_j \mid I_j) \geq g_j \geq 0$, and $w_j \geq 1 - \delta_j$, so $w_j h(E_j \mid I_j) \geq (1 - \delta_j) g_j$.

Adding (i)–(iii): the left side is at least $\sum_j [\delta_j + (1 - \delta_j)] g_j = \sum_j h(E_j \mid V_{j-1}) = h(V)$ by the chain rule and (F1). □

**Theorem 1** ($\ell_p$ Composition Inequality)**.** *Let $D$ be a database, $\mathcal{Q}(D) \neq \emptyset$, and let $\mu_p(F_j \mid I_j)$ be the $\ell_p$-norm of the degree sequence of $E_j$-values per $I_j$-value in a relation $R_j \supseteq \pi_{A_j} \mathcal{Q}(D)$—for instance the block join $\bowtie_{R \in F_j} R$ computed exactly on $D$. If $w$ satisfies (C), then*

$$|\mathcal{Q}(D)| \;\leq\; \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}.$$

*Proof.* Let $h$ be the entropy of the uniform distribution on $\mathcal{Q}(D)$; it is a polymatroid and $h(V) = \log_2 |\mathcal{Q}(D)|$. The marginal on $A_j$ is supported on $\pi_{A_j} \mathcal{Q}(D) \subseteq R_j$, so the $\ell_p$ degree inequality of Abo Khamis et al. (PODS 2024) gives, for $p \in [1, \infty]$, $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \leq \log_2 \mu_p(F_j \mid I_j)$. Multiply by $w_{j,p} \geq 0$, sum, and apply Lemma 3. □

**Scope.** The proof uses only monotonicity and submodularity, so it holds over the whole polymatroid cone and for every hyperpath, block size and $p \geq 1$. It requires (C) exactly: an LP solver may return weights violating (C) by about $10^{-7}$; scaling $w$ by $1/(1 - \varepsilon)$, for a coverage shortfall $\varepsilon = 1 - \min_a(\text{coverage of } a)$, restores (C) and raises a bound $B$ by the factor $B^{\varepsilon/(1-\varepsilon)}$, which grows with $B$. At the solver's feasibility tolerance ($\varepsilon \leq 10^{-7}$) and the largest cover bound we report ($B < 2^{100}$) the factor is below $1 + 10^{-5}$; the released code does not apply it, so reported bounds may sit below their rescaled values by at most that factor. It requires exact statistics of the database being bounded (or deterministic upper bounds on them, since the bound is monotone in each $\mu$); a sampled or sketched statistic, such as the Hutchinson estimate of the common-neighbour $\ell_2$ in the paper, is outside the guarantee. When some block join is empty the query is empty and the inequality holds trivially (the released LP gives such a factor weight zero, which only loosens the bound).

**Proposition 1** (Necessity)**.** *If nonnegative weights $(w_{j,p})$ make the product above an upper bound on $|\mathcal{Q}(D)|$ for every database $D$ whose atoms' relations are independent (no two atoms are copies of one relation), they satisfy (C). Hence on a fixed hyperpath the cover LP is optimal over all weightings of its statistics that are valid on every such database.*

*Proof.* Fix $a \in V$ and $N \geq 2$, and let $D_N$ give $a$ the domain $[N]$ and every other attribute a single value, each relation holding all tuples over these domains. Then $|\mathcal{Q}(D_N)| = N$, and the block join of $F_j$ has $N$ tuples if $a \in A_j$ and one otherwise, so $\mu_p(F_j \mid I_j)$ is $N^{1/p}$ if $a \in I_j$, $N$ if $a \in E_j$, and

1 if $a \notin A_j$. The bound is $N^{c_a}$ with $c_a$ the left side of (C) at $a$, and $N^{c_a} \geq N$ forces $c_a \geq 1$. (The entropy of $D_N$'s join is the modular polymatroid $h(S) = [a \in S] \log_2 N$, so the proposition also holds over the polymatroid cone.) □

**Self-joins.** The construction sets each atom's relation independently, which a self-join forbids: in $R(a,b) \bowtie R(b,c) \bowtie R(c,a)$ the three atoms share $R$, $D_N$ is not a database of that query, and indeed weight 3 on the first step's $|R|$ and 0 elsewhere gives the valid bound $|R|^3$ although $c$ is uncovered. Sufficiency (the Composition Inequality) is unaffected, since every self-join database is also a database of the independent query; necessity, and with it the reading that reweighting a fixed hyperpath's statistics is exhausted, holds for independent atoms only.

**Why the elementary argument does not suffice.** If $w_j \geq 1$ for every step, Lemma 3 follows from (iii) alone. The LP's optima on cyclic queries have $w_j < 1$ (e.g. 1/3 on the closing step of a cycle); the deficit $\delta_j$ is paid for by later interfaces, which is exactly step (ii). The cyclic improvement and the need for Lemma 2 have the same source.

**Independent checks.** An LP over the polymatroid cone decides Lemma 3 at fixed weights; enumerating every vertex of the feasible weight region decides it for all weights. Over 6,197 hyperpath structures on eleven shapes with $|V| \leq 6$ and block sizes 1–3 the minimum slack is 0 to solver precision; an adversarial search over random abstract hyperpath schemes with $|V| \leq 5$ found no violation; and over 207,000 database–block-size checks the bound never fell below the exact count (`scripts/shannon_check_wide.py`, `scripts/validity_search.py`).

## 2 The Cyclic Improvement theorem

Order the cycle $C_k$ as $R_1(a_1, a_2), \ldots, R_k(a_k, a_1)$ and extend in this order with single-relation blocks. Let $N_j = |R_j|$, $M_j$ the maximum degree and $\mu_p(R_j)$ the $\ell_p$-norm of the degree sequence of $R_j$ from $a_j$ to $a_{j+1}$.

**Structure.** Step 1 has interface $\emptyset$ and extension $\{a_1, a_2\}$: its degree sequence is the single value $N_1$. A middle step $2 \leq j \leq k-1$ has interface $\{a_j\}$, extension $\{a_{j+1}\}$ and $\mu_\infty = M_j$. The closing step $k$ has interface $\{a_k, a_1\}$ and empty extension; its conditional degrees are all 1, so $\mu_\infty = 1$ and $\mu_1 = N_k$. With $p = \infty$ and unit weights this gives $B_{\max} = N_1 \prod_{j=2}^{k-1} M_j$.

**A feasible point for every common $p \geq 1$.** Put weight $\frac{p}{p+1}$ with exponent $p$ on every middle step, $\frac{p}{p+1}$ on step 1 and $\frac{1}{p+1}$ on the closing step with exponent 1. The cover constraints read: $a_1$: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; $a_2$: $\frac{p}{p+1} + \frac{p/(p+1)}{p} = 1$; $a_i$ for $3 \leq i \leq k-1$: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; $a_k$: $\frac{p}{p+1} + \frac{1}{p+1} = 1$. All are tight, so the LP optimum satisfies

$$B_{\mathrm{LP}} \leq N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}} \quad \text{and} \quad \frac{B_{\max}}{B_{\mathrm{LP}}} \geq \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{p/(p+1)}}.$$

**The exponent must be common (or non-increasing).** The constraint of the attribute between middle edges $j$ and $j{+}1$ is $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \geq 1$, i.e. $p_{j+1} \leq p_j$. Choosing each edge's best exponent independently is therefore not feasible in general (e.g. $p_j = 1$, $p_{j+1} = \infty$ covers that attribute only to $\frac{1}{2}$); the LP may still mix exponents within a step.

**Growth.** At $p = 2$ a middle factor $M_j/\mu_2(R_j)^{2/3}$ exceeds 1 exactly when $\mu_2(R_j)^2 < M_j^3$: the sum of squared degrees is below the cube of the maximum degree. A few high-degree values satisfy it; many low-degree ones need not ($n$ values of uniform degree $d$ qualify iff $n < d$, and $10^4$ values of degree 1 plus one of degree 20 do not). When every middle factor is at least $c > 1$ the ratio grows geometrically in the number $k - 2$ of middle edges. As $p \to \infty$ every factor tends to 1, and LpH-CEG returns $\min(B_{\max}, B_{\mathrm{LP}})$, so the realized ratio is always at least 1. The boundary factor $(N_1/N_k)^{1/(p+1)}$ does not depend on $k$; it is 1 for a uniform cycle.

**Acyclic queries.** On a star the cover constraints force unit weight on every step and the LP returns the max-degree bound. On a path the LP can use an interface at $p = 1$ in place of an extension (on a 4-path it returns weights $(1, 0, 1)$), so acyclicity alone does not force unit weights; what redundant coverage on cycles adds is the slack exploited above.

## 3 A spectral reading of the common-neighbour statistic (post hoc)

For an even $k$-cycle of binary relations the count is a trace, $|\mathcal{Q}| = \mathrm{Tr}(A_1 \cdots A_k)$ for the incidence matrices $A_i$ ($v_i \times v_{i+1}$), and the common-neighbour $\ell_2$ norm is the Frobenius (Schatten-2) norm of the block product $B = A_i A_{i+1}$. Hölder's inequality for Schatten norms gives, over $m = k/2$ consecutive blocks, $|\mathcal{Q}| \le \prod_j \|B_j\|_{S_m}$, and $\|B\|_{S_m}^m \le \sigma_{\max}(B)^{m-2}\|B\|_F^2$ with $\sigma_{\max}$ bounded above by a Collatz–Wielandt certificate on $B^\top B$. Two numbers per block, both from the pair join ($\sigma_{\max}$ lies outside the polymatroid catalog), give a bound tighter than the polymatroid program on 96/240 long-suite queries (by a median 1.25× there, though 1.11× looser over all) and never below an exact count; with each block's top 24 singular values (Lanczos; uncertified, yet above every exact count) it is tighter on 220/240 and takes IMDb at $|V|$=10 from 14.9× to 3.2× the truth. The next step on the statistics axis may thus be spectral order rather than block size.

## 4 Additional $bs = 1$ experiments: JOB, TPC-H-like, data size and skew

On the JOB-medium subset (CI: 3K, MK: 2K, MC: 1K rows) the two co-actor 4-cycles improve 1.94× (with keywords) and 2.80× (with companies) at $bs = 1$ with weights $w$=0.5, and a star query by exactly 1.00×; the IMDb suite of the paper is the systematic version. On a small TPC-H-like synthetic schema (SF=0.01; not `dbgen` data) every cyclic query we built yields 1.00×; LpH-CEG needs *cyclic structure* together with *degree skew* in a middle relation. On synthetic cycle-4 (10 seeds) the improvement grows with data size, from 5.9±0.7× at 100 vertices to 33.9±1.5× at 2K (relative std <12%; a single-run sweep reaches 57× at 5K), and peaks at moderate skew: 2.3±0.5× at $\alpha$=0.5, 16.1±0.8× at $\alpha$=2.0, 6.6±0.4× at $\alpha$=3.0. At low skew all $\ell_p$-norms nearly coincide, at extreme skew $\ell_\infty$ is already near-tight, and in between $\ell_2$ captures the heavy tail.

Every number in this section is produced by the scripts in the repository and checked by its claim audit.

## 5 Multi-way ordering by per-step residuals

The per-step residuals can order a worst-case-optimal join greedily. As implemented, the residual of a variable prefix is the LP re-solved on the query with the prefix attributes projected out, which bounds the residual for every value of the prefix (and could be computed for LpBound in the

same way): GenericJoin iterations drop 2.3× on the Hetionet cycle-4 in our simulator, but over 66 synthetic queries the bound-guided order is 1.3% worse at the median (20 wins, 43 losses; Wilcoxon $p$=0.003, indicative, as queries share shapes and seeds), so it is a targeted heuristic.

## 6 Complexity

Collecting $\mu_p(F \mid I)$ for a single relation is one scan, $O(N)$; for a two-relation block it materializes the pair join, except for the interfaces computable from single-relation degree tables (all but the common-neighbour interface of a binary block; paragraph "Which statistic needs the join" of the paper). The cover LP has $m|\mathcal{P}|$ variables and $|V|$ constraints; it solves in milliseconds for the queries in the paper. The flow LP over a $bs \leq 2$ catalog is polynomial in the number of constraints and solves in 0.03–4.4 s at $k = 3$–16 over $4k$ orders.